CPSC 536C: Algorithms for Convex Optimization
Open Problems

Department of Computer Science
University of British Columbia

Winter I, 2026-2027

Open Projects and Student Lectures

- Minimax Theorems and Game Theory
- Efficient Convex Oracles and Reductions
  https://arxiv.org/abs/1706.07357
- Shallow-Cut Barrier Method (Week 4-5)
  Reference [Grotschel, Lovasz, Schrijver 3.3, 4.3]
- Geometric Descent (Week 5-7)
  https://arxiv.org/abs/1506.08187
- Vaidya + Lee-Sidford-Wong Cutting Plane Method (Week 4-5)
  https://arxiv.org/abs/1508.04874
- Stochastic Gradient Descent
- Krylov/ Preconditioning for Linear Systems
- Smoothing for Lower bounds (Week 5-7)
  https://www.sciencedirect.com/science/article/pii/S0885064X14000831
- Smoothing for Acceleration (Week 5-7)
  https://link.springer.com/article/10.1007/s10107-004-0552-5
- Performance Estimation (SDP) Framework (Week 5-8)
  https://link.springer.com/article/10.1007/s10107-013-0653-0
- Accelerated Gradient Descent
  https://www.mathnet.ru/php/archive.phtml?wshow=paper&jrnid=dan&paperid=46009&option_lang=eng, [Nesterov 2.2]
- Acceleration via Linear coupling (Week 7-9)
  https://arxiv.org/abs/1407.1537
- Acceleration via Approx Duality gap (Week 7-9)
  https://epubs.siam.org/doi/10.1137/18M1172314)

- Multiplicative Weight Update Method
  https://www.cs.princeton.edu/~arora/pubs/MWsurvey.pdf
- Laplacian Solvers (Week 7-10)
  https://dl.acm.org/doi/abs/10.1145/1007352.1007372 ...
- Max Flow
  https://www.cs.cmu.edu/~15859n/RelatedWork/LeeRaoStrivastava.pdf
- Generalized GD for Max Flow
  https://epubs.siam.org/doi/10.1137/1.9781611973402.16
- Spectral Sparsification
  https://arxiv.org/abs/1506.04838
- Learning Rate for Online Optimization (Week 7-10)
  https://proceedings.mlr.press/v119/fang20a.html
- Relative strong convexity/ smoothness (Week 7-10)
  https://epubs.siam.org/doi/10.1137/16M1099546
- Vaidya + Lee-Sidford Barrier (Week 10-12)
  https://ieeexplore.ieee.org/document/63500, https://arxiv.org/abs/1910.08033
- Universal Barrier
  (Nesterov, Nemirovski 2.5)
- Entropic Barrier (Week 10-12)
  https://arxiv.org/abs/1412.1587
- Predictor/Corrector Method for LP (Week 11-13)
  https://www.jstor.org/stable/3690133
- Straight-Line Complexity (Week 11-13)
  https://arxiv.org/abs/2206.08810

Instructions

- Choose topic from list or propose own topic (with connection to optimization); send me an email with topic by 3rd week of class
- I will decide schedule by 4th week of class;
- Send me an outline of your talk by 2 office hours before your talk; come to office hours if you have any questions or want to go over the outline; I will send comments about what should be covered in the lecture;
- Make an overleaf (or github or other shared document) and share the outline with the scribe for your lecture.
- For scribing: send me pdf of rough outline by 1.5 weeks after the lecture
- I will send comments and you will have another week to finish/ polish before we post on the class website

Student Lecture (Tentative) Schedule

- Minimax (October 5): Irene, Jeffrey
- SGD (October 21): Oluwanifemi
- PEPS (October 21): Jason
- Krylov/Preconditioning/CG (October 26): William
- Accelerated GD (October 28): Jan, Tung
- Max flow (November 2): Efe
- Laplacian Solvers (November 4): Siyuan
- Frank-Wolfe (November 16): Alexandre
- MWU, Online Learning (November 18 or 23): Towa, Aaron
- Relative Smoothness/ Modular Duality (November 25): Chia-Hong, Michael
- Entropic Barrier (December 7): Zachary

Student Lecture Schedule (WII 2025-2026)

- Stochastic Gradient Descent (Lecturer: Kevin, Feb 9; Scribe: Arqam)
- Conjugate Gradient (Lecturer: Chen Greif, Feb 23; Scribe: Kevin)
- Max Flow (Lecturer: Hasti, Feb 25; Scribe: Mitali)
- Spectral Sparsification (Lecturer: Tong, Mar 2; Scribe: Vasiliki)
- Multiplicative Weight Update Method (Lecturers: Inzaghi + Yin, March 4; Scribe: Tong)
- Accelerated Gradient Descent (Lecturer: Trevor, March 11; Scribe: Hasti)
- Spectral Descent Algorithm (Lecturer: Ying Qi, March 16; Scribe: Inzaghi + Yin)
- Universal Barrier (Lecturer: Yixian, March 30; Scribe: Ying Qi)
- Minimax Theorems and Game Theory (Lecturers: Vasiliki + Mitali, April 6-8; Scribe: Trevor)
- Efficient Convex Oracles and Reductions (Lecturer: Arqam, April 6-8; Scribe: Yixian)